# Final Results

| Model type | Training images | Best epoch | Validation accuracy at best epoch | Test accuracy |
|---|---|---|---|---|
| From scratch | 1,000 | 38 | 75.800% | 74.600% |
| From scratch | 1,500 | 46 | 82.400% | 81.400% |
| From scratch | 2,000 | 47 | 85.400% | 82.600% |
| Pretrained VGG16 | 1,000 | 25 | 99.000% | 97.800% |
| Pretrained VGG16 | 1,500 | 11 | 98.800% | 97.000% |
| Pretrained VGG16 | 2,000 | 30 | 99.000% | 97.600% |

## Summary

The pretrained VGG16 models achieved substantially higher validation and test accuracy than the models trained from scratch. The custom CNN's test accuracy improved from 74.6% to 82.6% as the training set increased from 1,000 to 2,000 images. The pretrained models achieved test accuracies between 97.0% and 97.8%, with the model trained on 1,000 images producing the highest test accuracy. Overall accuarcy between the three pretrained models is all within .8%, which could be due to ordinary sampling variation. All models were optimized using dropout and data augmentation techniques.

In [6]:
```
from pathlib import Path

project_dir = Path.home() / "ml" / "cats_dogs"
print("Project folder exists:", project_dir.exists())
print("Files in folder:")
for item in project_dir.iterdir():
    print(item.name)
```

```
Project folder exists: True
Files in folder:
cats_dogs_assignment
cats_vs_dogs_small.zip
cats_vs_dogs_small.zip:Zone.Identifier
.ipynb_checkpoints
Untitled.ipynb
cats_vs_dogs_small
```

In [7]:
```
import os
import shutil
import pathlib
import zipfile
```

```
project_dir = pathlib.Path.home() / "ml" / "cats_dogs"
zip_path = project_dir / "cats_vs_dogs_small.zip"
original_dir = project_dir / "cats_vs_dogs_small"
new_base_dir = project_dir / "cats_dogs_assignment"
```

In [8]:
```
from pathlib import Path
import zipfile

project_dir = Path.home() / "ml" / "cats_dogs"
zip_path = project_dir / "cats_vs_dogs_small.zip"
original_dir = project_dir / "cats_vs_dogs_small"

if not original_dir.exists():
    with zipfile.ZipFile(zip_path, "r") as z:
        z.extractall(project_dir)

print("Extraction complete.")
print("Project contents:", [p.name for p in project_dir.iterdir()])
```

```
Extraction complete.
Project contents: ['cats_dogs_assignment', 'cats_vs_dogs_small.zip', 'cats_vs_dogs_s
mall.zip:Zone.Identifier', '.ipynb_checkpoints', 'Untitled.ipynb', 'cats_vs_dogs_sma
ll']
```

In [9]:
```
for path in sorted(original_dir.iterdir()):
    if path.is_dir():
        print(f"\n{path.name}/")
        for subdir in sorted(path.iterdir()):
            if subdir.is_dir():
                print(f"  {subdir.name}: {len(list(subdir.glob('*.jpg')))} JPG imag
```

```
test/
  cats: 500 JPG images
  dogs: 500 JPG images

train/
  cats: 1000 JPG images
  dogs: 1000 JPG images

validation/
  cats: 500 JPG images
  dogs: 500 JPG images
```

In [10]:
```
import pathlib
import shutil

project_dir = pathlib.Path.home() / "ml" / "cats_dogs"
original_dir = project_dir / "cats_vs_dogs_small"
new_base_dir = project_dir / "cats_dogs_assignment"

def make_subset(source_split, subset_name, end_index):
    for category in ("cats", "dogs"):
        source_dir = original_dir / source_split / category
        destination_dir = new_base_dir / subset_name / category
        destination_dir.mkdir(parents=True, exist_ok=True)
```

```
        fnames = sorted(source_dir.glob("*.jpg"))[:end_index]

        for fname in fnames:
            shutil.copy2(fname, destination_dir / fname.name)

make_subset("train", "train_1000", 500)
make_subset("train", "train_1500", 750)
make_subset("train", "train_2000", 1000)
make_subset("validation", "validation", 250)
make_subset("test", "test", 250)

print("Subsets created under:", new_base_dir)
```

```
Subsets created under: /home/claxta11/ml/cats_dogs/cats_dogs_assignment
```

In [11]:
```
for split in ("train_1000", "train_1500", "train_2000",
              "validation", "test"):
    print(f"\n{split}")
    for category in ("cats", "dogs"):
        folder = new_base_dir / split / category
        print(f"  {category}: {len(list(folder.glob('*.jpg')))} images")
```

```
train_1000
  cats: 500 images
  dogs: 500 images

train_1500
  cats: 750 images
  dogs: 750 images

train_2000
  cats: 1000 images
  dogs: 1000 images

validation
  cats: 250 images
  dogs: 250 images

test
  cats: 250 images
  dogs: 250 images
```

In [12]:
```
import tensorflow as tf

print("TensorFlow:", tf.__version__)
print("Available GPUs:", tf.config.list_physical_devices("GPU"))
```

```
TensorFlow: 2.20.0
Available GPUs: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
```

In [13]:
```
#instantiate model_from_scratch
from tensorflow import keras
from tensorflow.keras import layers

#Adding data augmentation to scratch model
data_augmentation = keras.Sequential(
    [
```

```
        layers.RandomFlip("horizontal"),
        layers.RandomRotation(0.1),
        layers.RandomZoom(0.2),
    ]
)


#function to make the model from scratch and configure
def make_model_from_scratch():
    inputs = keras.Input(shape=(180, 180, 3))
    x = data_augmentation(inputs)
    x = layers.Rescaling(1./255)(x)
    x = layers.Conv2D(filters=32, kernel_size=3, activation="relu")(x)
    x = layers.MaxPooling2D(pool_size=2)(x)
    x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(x)
    x = layers.MaxPooling2D(pool_size=2)(x)
    x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
    x = layers.MaxPooling2D(pool_size=2)(x)
    x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
    x = layers.MaxPooling2D(pool_size=2)(x)
    x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
    x = layers.Flatten()(x)
    x = layers.Dropout(0.5)(x)
    outputs = layers.Dense(1, activation="sigmoid")(x)

    model = keras.Model(inputs=inputs, outputs=outputs)

    model.compile(
        loss="binary_crossentropy",
        optimizer="rmsprop",
        metrics=["accuracy"]
    )

    return model
```

In [14]:

```
#reading images
from tensorflow.keras.utils import image_dataset_from_directory

train_1000_dataset = image_dataset_from_directory(
    new_base_dir / "train_1000",
    image_size=(180, 180),
    batch_size=32
)

train_1500_dataset = image_dataset_from_directory(
    new_base_dir / "train_1500",
    image_size=(180, 180),
    batch_size=32
)

train_2000_dataset = image_dataset_from_directory(
    new_base_dir / "train_2000",
    image_size=(180, 180),
    batch_size=32
)
```

```
validation_dataset = image_dataset_from_directory(
    new_base_dir / "validation",
    image_size=(180, 180),
    batch_size=32
)

test_dataset = image_dataset_from_directory(
    new_base_dir / "test",
    image_size=(180, 180),
    batch_size=32
)
```

```
Found 1000 files belonging to 2 classes.
Found 1500 files belonging to 2 classes.
Found 2000 files belonging to 2 classes.
Found 500 files belonging to 2 classes.
Found 500 files belonging to 2 classes.
```

In [15]:
```
#Make three distinct models
model_1000 = make_model_from_scratch()
model_1500 = make_model_from_scratch()
model_2000 = make_model_from_scratch()

#function for callbacks
def make_callbacks(training_size):
    return [
        keras.callbacks.ModelCheckpoint(
            filepath=f"convnet_from_scratch_{training_size}.keras",
            save_best_only=True,
            monitor="val_loss"
        )
    ]
```

In [16]:
```
history_1000 = model_1000.fit(
    train_1000_dataset,
    epochs=100,
    validation_data=validation_dataset,
    callbacks=make_callbacks("1000")
)

history_1500 = model_1500.fit(
    train_1500_dataset,
    epochs=100,
    validation_data=validation_dataset,
    callbacks=make_callbacks("1500")
)

history_2000 = model_2000.fit(
    train_2000_dataset,
    epochs=100,
    validation_data=validation_dataset,
    callbacks=make_callbacks("2000")
)


test_model_1000 = keras.models.load_model(
```

```
    "convnet_from_scratch_1000.keras")
test_loss_1000, test_acc_1000 = test_model_1000.evaluate(test_dataset)
print(f"1000 Test accuracy: {test_acc_1000:.3f}")

test_model_1500 = keras.models.load_model(
    "convnet_from_scratch_1500.keras")
test_loss_1500, test_acc_1500 = test_model_1500.evaluate(test_dataset)
print(f"1500 Test accuracy: {test_acc_1500:.3f}")

test_model_2000 = keras.models.load_model(
    "convnet_from_scratch_2000.keras")
test_loss_2000, test_acc_2000 = test_model_2000.evaluate(test_dataset)
print(f"2000 Test accuracy: {test_acc_2000:.3f}")
```

```
Epoch 1/100
/home/claxta11/miniconda3/envs/tensorflow/lib/python3.12/site-packages/keras/src/trainers/epoch_iterator.py:74: UserWarning: `shuffle=True` was passed, but will be ignored since the data `x` was provided as a tf.data.Dataset. The Dataset is expected to already be shuffled (via `.shuffle(buffer_size)`).
  self.data_adapter = data_adapters.get_data_adapter(
2026-10-15 19:06:37.496529: I external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:473] Loaded cuDNN version 92600
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 8s 94ms/step - accuracy: 0.5140 - loss: 0.7079 - val_accu
racy: 0.5000 - val_loss: 0.6928
Epoch 2/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.4840 - loss: 0.6935 - val_accu
racy: 0.5000 - val_loss: 0.6919
Epoch 3/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 56ms/step - accuracy: 0.5220 - loss: 0.6932 - val_accu
racy: 0.5000 - val_loss: 0.6897
Epoch 4/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.5670 - loss: 0.6900 - val_accu
racy: 0.6200 - val_loss: 0.6817
Epoch 5/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.5670 - loss: 0.6828 - val_accu
racy: 0.6080 - val_loss: 0.6704
Epoch 6/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.5760 - loss: 0.6846 - val_accu
racy: 0.5180 - val_loss: 0.6724
Epoch 7/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.5990 - loss: 0.6592 - val_accu
racy: 0.5020 - val_loss: 0.7094
Epoch 8/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.6050 - loss: 0.6625 - val_accu
racy: 0.5000 - val_loss: 0.8140
Epoch 9/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.6180 - loss: 0.6560 - val_accu
racy: 0.5060 - val_loss: 0.8474
Epoch 10/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.6140 - loss: 0.6481 - val_accu
racy: 0.6340 - val_loss: 0.6425
Epoch 11/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.6750 - loss: 0.6158 - val_accu
racy: 0.5800 - val_loss: 0.7692
Epoch 12/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 56ms/step - accuracy: 0.6540 - loss: 0.6140 - val_accu
racy: 0.6560 - val_loss: 0.6149
Epoch 13/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.6600 - loss: 0.6265 - val_accu
racy: 0.6360 - val_loss: 0.6196
Epoch 14/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.6800 - loss: 0.6057 - val_accu
racy: 0.6760 - val_loss: 0.6105
Epoch 15/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.6840 - loss: 0.6086 - val_accu
racy: 0.7180 - val_loss: 0.5698
Epoch 16/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.6900 - loss: 0.5848 - val_accu
racy: 0.7140 - val_loss: 0.5583
Epoch 17/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.6910 - loss: 0.5678 - val_accu
racy: 0.6440 - val_loss: 0.7109
Epoch 18/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7110 - loss: 0.5598 - val_accu
racy: 0.5120 - val_loss: 1.8046
Epoch 19/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.6990 - loss: 0.5909 - val_accu
racy: 0.6480 - val_loss: 0.7883
```

```
Epoch 20/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7140 - loss: 0.5573 - val_accu
racy: 0.6340 - val_loss: 0.7428
Epoch 21/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7080 - loss: 0.6058 - val_accu
racy: 0.6840 - val_loss: 0.6088
Epoch 22/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.7280 - loss: 0.5302 - val_accu
racy: 0.5760 - val_loss: 1.1442
Epoch 23/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7310 - loss: 0.5569 - val_accu
racy: 0.7200 - val_loss: 0.6024
Epoch 24/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 55ms/step - accuracy: 0.7370 - loss: 0.5610 - val_accu
racy: 0.7260 - val_loss: 0.5212
Epoch 25/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7260 - loss: 0.5148 - val_accu
racy: 0.7000 - val_loss: 0.6040
Epoch 26/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7450 - loss: 0.5210 - val_accu
racy: 0.6880 - val_loss: 0.5911
Epoch 27/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7300 - loss: 0.5328 - val_accu
racy: 0.6980 - val_loss: 0.6654
Epoch 28/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.7620 - loss: 0.4994 - val_accu
racy: 0.7420 - val_loss: 0.5009
Epoch 29/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7560 - loss: 0.4990 - val_accu
racy: 0.7380 - val_loss: 0.5452
Epoch 30/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7780 - loss: 0.4657 - val_accu
racy: 0.6900 - val_loss: 0.7323
Epoch 31/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7710 - loss: 0.4918 - val_accu
racy: 0.7360 - val_loss: 0.5965
Epoch 32/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7600 - loss: 0.4845 - val_accu
racy: 0.7340 - val_loss: 0.6015
Epoch 33/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7940 - loss: 0.4572 - val_accu
racy: 0.7600 - val_loss: 0.5529
Epoch 34/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.7850 - loss: 0.4534 - val_accu
racy: 0.7260 - val_loss: 0.5722
Epoch 35/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7870 - loss: 0.4574 - val_accu
racy: 0.6780 - val_loss: 0.8639
Epoch 36/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.7980 - loss: 0.4299 - val_accu
racy: 0.7500 - val_loss: 0.6531
Epoch 37/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.7820 - loss: 0.4368 - val_accu
racy: 0.7260 - val_loss: 0.5703
Epoch 38/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 54ms/step - accuracy: 0.7950 - loss: 0.4437 - val_accu
```

```
racy: 0.7580 - val_loss: 0.4968
Epoch 39/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8250 - loss: 0.4005 - val_accu
racy: 0.7660 - val_loss: 0.5900
Epoch 40/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8210 - loss: 0.3996 - val_accu
racy: 0.6220 - val_loss: 0.7766
Epoch 41/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8030 - loss: 0.4144 - val_accu
racy: 0.7260 - val_loss: 0.6358
Epoch 42/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8280 - loss: 0.3864 - val_accu
racy: 0.7340 - val_loss: 0.6593
Epoch 43/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8200 - loss: 0.3852 - val_accu
racy: 0.7700 - val_loss: 0.6015
Epoch 44/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8470 - loss: 0.3650 - val_accu
racy: 0.7700 - val_loss: 0.6080
Epoch 45/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8340 - loss: 0.3829 - val_accu
racy: 0.7780 - val_loss: 0.5726
Epoch 46/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8400 - loss: 0.3667 - val_accu
racy: 0.7960 - val_loss: 0.5698
Epoch 47/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8260 - loss: 0.3851 - val_accu
racy: 0.7360 - val_loss: 0.6011
Epoch 48/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8330 - loss: 0.3513 - val_accu
racy: 0.7540 - val_loss: 0.5477
Epoch 49/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8650 - loss: 0.3270 - val_accu
racy: 0.7420 - val_loss: 0.9073
Epoch 50/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8410 - loss: 0.3619 - val_accu
racy: 0.7640 - val_loss: 0.6696
Epoch 51/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8470 - loss: 0.3320 - val_accu
racy: 0.7800 - val_loss: 0.5496
Epoch 52/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8490 - loss: 0.3297 - val_accu
racy: 0.7480 - val_loss: 0.7045
Epoch 53/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8640 - loss: 0.2947 - val_accu
racy: 0.7760 - val_loss: 0.6833
Epoch 54/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8560 - loss: 0.3254 - val_accu
racy: 0.7020 - val_loss: 1.0663
Epoch 55/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8830 - loss: 0.2838 - val_accu
racy: 0.7540 - val_loss: 0.6844
Epoch 56/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8660 - loss: 0.3045 - val_accu
racy: 0.7640 - val_loss: 0.6200
Epoch 57/100
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32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8830 - loss: 0.3014 - val_accu
racy: 0.7860 - val_loss: 0.6678
Epoch 58/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8830 - loss: 0.2741 - val_accu
racy: 0.7040 - val_loss: 0.8183
Epoch 59/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8690 - loss: 0.3066 - val_accu
racy: 0.7520 - val_loss: 0.8148
Epoch 60/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8560 - loss: 0.3200 - val_accu
racy: 0.7700 - val_loss: 0.6344
Epoch 61/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8760 - loss: 0.2732 - val_accu
racy: 0.7880 - val_loss: 0.5942
Epoch 62/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8960 - loss: 0.2433 - val_accu
racy: 0.7820 - val_loss: 0.7008
Epoch 63/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8740 - loss: 0.2997 - val_accu
racy: 0.7700 - val_loss: 0.6637
Epoch 64/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8910 - loss: 0.2700 - val_accu
racy: 0.7840 - val_loss: 0.6611
Epoch 65/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9030 - loss: 0.2297 - val_accu
racy: 0.7620 - val_loss: 0.8451
Epoch 66/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8880 - loss: 0.2748 - val_accu
racy: 0.7740 - val_loss: 0.7823
Epoch 67/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9000 - loss: 0.2542 - val_accu
racy: 0.7780 - val_loss: 0.6360
Epoch 68/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.8970 - loss: 0.2501 - val_accu
racy: 0.8060 - val_loss: 0.6704
Epoch 69/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9100 - loss: 0.2276 - val_accu
racy: 0.7880 - val_loss: 0.8228
Epoch 70/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9160 - loss: 0.2235 - val_accu
racy: 0.7840 - val_loss: 0.6505
Epoch 71/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9010 - loss: 0.2380 - val_accu
racy: 0.7540 - val_loss: 0.7699
Epoch 72/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.8930 - loss: 0.2572 - val_accu
racy: 0.7920 - val_loss: 0.6786
Epoch 73/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9160 - loss: 0.2150 - val_accu
racy: 0.7700 - val_loss: 0.7300
Epoch 74/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9310 - loss: 0.1973 - val_accu
racy: 0.7800 - val_loss: 0.9303
Epoch 75/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9120 - loss: 0.2387 - val_accu
racy: 0.7580 - val_loss: 0.7411
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Epoch 76/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9200 - loss: 0.1995 - val_accu
racy: 0.7780 - val_loss: 0.8480
Epoch 77/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9150 - loss: 0.2275 - val_accu
racy: 0.7920 - val_loss: 0.6558
Epoch 78/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9160 - loss: 0.1960 - val_accu
racy: 0.7940 - val_loss: 0.7102
Epoch 79/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9130 - loss: 0.2441 - val_accu
racy: 0.7840 - val_loss: 0.8170
Epoch 80/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9360 - loss: 0.1662 - val_accu
racy: 0.8060 - val_loss: 0.7212
Epoch 81/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9230 - loss: 0.2134 - val_accu
racy: 0.7820 - val_loss: 0.7789
Epoch 82/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9160 - loss: 0.2163 - val_accu
racy: 0.7760 - val_loss: 1.0218
Epoch 83/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9080 - loss: 0.2275 - val_accu
racy: 0.7820 - val_loss: 0.8017
Epoch 84/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9260 - loss: 0.1915 - val_accu
racy: 0.7940 - val_loss: 0.7955
Epoch 85/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9200 - loss: 0.1938 - val_accu
racy: 0.7560 - val_loss: 1.0570
Epoch 86/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9320 - loss: 0.1686 - val_accu
racy: 0.7900 - val_loss: 0.7562
Epoch 87/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9290 - loss: 0.1689 - val_accu
racy: 0.7980 - val_loss: 0.8146
Epoch 88/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9230 - loss: 0.2147 - val_accu
racy: 0.8000 - val_loss: 0.7415
Epoch 89/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.9450 - loss: 0.1684 - val_accu
racy: 0.7540 - val_loss: 0.8511
Epoch 90/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9410 - loss: 0.1574 - val_accu
racy: 0.7720 - val_loss: 1.1969
Epoch 91/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.9420 - loss: 0.1457 - val_accu
racy: 0.7920 - val_loss: 0.8799
Epoch 92/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.9280 - loss: 0.1697 - val_accu
racy: 0.7880 - val_loss: 1.0804
Epoch 93/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9540 - loss: 0.1519 - val_accu
racy: 0.7920 - val_loss: 0.8091
Epoch 94/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9340 - loss: 0.1619 - val_accu
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racy: 0.8080 - val_loss: 0.8947
Epoch 95/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9290 - loss: 0.1780 - val_accu
racy: 0.7920 - val_loss: 0.8098
Epoch 96/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9420 - loss: 0.1472 - val_accu
racy: 0.7760 - val_loss: 0.9518
Epoch 97/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9380 - loss: 0.1522 - val_accu
racy: 0.7880 - val_loss: 0.8541
Epoch 98/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9520 - loss: 0.1441 - val_accu
racy: 0.7660 - val_loss: 1.1628
Epoch 99/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 52ms/step - accuracy: 0.9500 - loss: 0.1402 - val_accu
racy: 0.7700 - val_loss: 1.1050
Epoch 100/100
32/32 ━━━━━━━━━━━━━━━━━━━━ 2s 53ms/step - accuracy: 0.9510 - loss: 0.1236 - val_accu
racy: 0.7940 - val_loss: 1.1101
Epoch 1/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 4s 77ms/step - accuracy: 0.4913 - loss: 0.7031 - val_accu
racy: 0.5000 - val_loss: 0.6926
Epoch 2/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.5347 - loss: 0.6941 - val_accu
racy: 0.5000 - val_loss: 0.6932
Epoch 3/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.5500 - loss: 0.6901 - val_accu
racy: 0.6100 - val_loss: 0.6729
Epoch 4/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.5860 - loss: 0.6778 - val_accu
racy: 0.6040 - val_loss: 0.6671
Epoch 5/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.6007 - loss: 0.6670 - val_accu
racy: 0.6020 - val_loss: 0.6410
Epoch 6/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.6220 - loss: 0.6559 - val_accu
racy: 0.6480 - val_loss: 0.6284
Epoch 7/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.6320 - loss: 0.6494 - val_accu
racy: 0.6000 - val_loss: 0.6580
Epoch 8/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.6580 - loss: 0.6325 - val_accu
racy: 0.6200 - val_loss: 0.6431
Epoch 9/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.6620 - loss: 0.6243 - val_accu
racy: 0.6680 - val_loss: 0.6056
Epoch 10/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.6800 - loss: 0.6089 - val_accu
racy: 0.6620 - val_loss: 0.5985
Epoch 11/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.6747 - loss: 0.6064 - val_accu
racy: 0.7100 - val_loss: 0.5407
Epoch 12/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7073 - loss: 0.5752 - val_accu
racy: 0.7060 - val_loss: 0.5475
Epoch 13/100
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47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7060 - loss: 0.5794 - val_accu
racy: 0.7540 - val_loss: 0.5077
Epoch 14/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.6947 - loss: 0.5776 - val_accu
racy: 0.7340 - val_loss: 0.5377
Epoch 15/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.7453 - loss: 0.5340 - val_accu
racy: 0.7300 - val_loss: 0.5257
Epoch 16/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7260 - loss: 0.5433 - val_accu
racy: 0.7620 - val_loss: 0.5032
Epoch 17/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.7413 - loss: 0.5182 - val_accu
racy: 0.7220 - val_loss: 0.5705
Epoch 18/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7480 - loss: 0.5143 - val_accu
racy: 0.7820 - val_loss: 0.4700
Epoch 19/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.7540 - loss: 0.5162 - val_accu
racy: 0.7440 - val_loss: 0.4974
Epoch 20/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.7527 - loss: 0.4971 - val_accu
racy: 0.7820 - val_loss: 0.4655
Epoch 21/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7687 - loss: 0.4870 - val_accu
racy: 0.7320 - val_loss: 0.5214
Epoch 22/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7673 - loss: 0.4914 - val_accu
racy: 0.7720 - val_loss: 0.4781
Epoch 23/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7727 - loss: 0.4739 - val_accu
racy: 0.7800 - val_loss: 0.4851
Epoch 24/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7867 - loss: 0.4591 - val_accu
racy: 0.7140 - val_loss: 0.5412
Epoch 25/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.7853 - loss: 0.4694 - val_accu
racy: 0.8100 - val_loss: 0.4445
Epoch 26/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7967 - loss: 0.4485 - val_accu
racy: 0.7800 - val_loss: 0.4685
Epoch 27/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.7813 - loss: 0.4505 - val_accu
racy: 0.7460 - val_loss: 0.5576
Epoch 28/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.7873 - loss: 0.4520 - val_accu
racy: 0.7800 - val_loss: 0.4408
Epoch 29/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8033 - loss: 0.4224 - val_accu
racy: 0.8040 - val_loss: 0.4513
Epoch 30/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8107 - loss: 0.4089 - val_accu
racy: 0.8000 - val_loss: 0.4417
Epoch 31/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8093 - loss: 0.4128 - val_accu
racy: 0.7840 - val_loss: 0.4656
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Epoch 32/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8020 - loss: 0.4129 - val_accu
racy: 0.7120 - val_loss: 0.6260
Epoch 33/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8200 - loss: 0.4149 - val_accu
racy: 0.7920 - val_loss: 0.4659
Epoch 34/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.8067 - loss: 0.4147 - val_accu
racy: 0.8240 - val_loss: 0.3952
Epoch 35/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 48ms/step - accuracy: 0.8327 - loss: 0.3807 - val_accu
racy: 0.7840 - val_loss: 0.5118
Epoch 36/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 48ms/step - accuracy: 0.8213 - loss: 0.3842 - val_accu
racy: 0.8180 - val_loss: 0.4313
Epoch 37/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8253 - loss: 0.3801 - val_accu
racy: 0.7980 - val_loss: 0.4738
Epoch 38/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8333 - loss: 0.3691 - val_accu
racy: 0.7940 - val_loss: 0.5451
Epoch 39/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8420 - loss: 0.3533 - val_accu
racy: 0.7900 - val_loss: 0.4313
Epoch 40/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8513 - loss: 0.3505 - val_accu
racy: 0.7620 - val_loss: 0.5798
Epoch 41/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8420 - loss: 0.3512 - val_accu
racy: 0.8080 - val_loss: 0.4659
Epoch 42/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8473 - loss: 0.3555 - val_accu
racy: 0.7820 - val_loss: 0.4793
Epoch 43/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8600 - loss: 0.3300 - val_accu
racy: 0.8180 - val_loss: 0.4593
Epoch 44/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8620 - loss: 0.3250 - val_accu
racy: 0.8260 - val_loss: 0.3991
Epoch 45/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8680 - loss: 0.3059 - val_accu
racy: 0.8180 - val_loss: 0.4571
Epoch 46/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 51ms/step - accuracy: 0.8560 - loss: 0.3315 - val_accu
racy: 0.8240 - val_loss: 0.3817
Epoch 47/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8727 - loss: 0.3036 - val_accu
racy: 0.7960 - val_loss: 0.5729
Epoch 48/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8680 - loss: 0.3020 - val_accu
racy: 0.8100 - val_loss: 0.4705
Epoch 49/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8780 - loss: 0.3176 - val_accu
racy: 0.8280 - val_loss: 0.3939
Epoch 50/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8880 - loss: 0.2737 - val_accu
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racy: 0.7980 - val_loss: 0.5381
Epoch 51/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8853 - loss: 0.2808 - val_accu
racy: 0.8440 - val_loss: 0.4355
Epoch 52/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8727 - loss: 0.2825 - val_accu
racy: 0.8140 - val_loss: 0.5119
Epoch 53/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8960 - loss: 0.2514 - val_accu
racy: 0.7760 - val_loss: 0.7451
Epoch 54/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8833 - loss: 0.2894 - val_accu
racy: 0.8280 - val_loss: 0.4468
Epoch 55/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8900 - loss: 0.2571 - val_accu
racy: 0.8040 - val_loss: 0.5635
Epoch 56/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8960 - loss: 0.2628 - val_accu
racy: 0.8340 - val_loss: 0.4743
Epoch 57/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8987 - loss: 0.2375 - val_accu
racy: 0.8160 - val_loss: 0.5341
Epoch 58/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8960 - loss: 0.2642 - val_accu
racy: 0.8000 - val_loss: 0.5676
Epoch 59/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.8973 - loss: 0.2414 - val_accu
racy: 0.8220 - val_loss: 0.5462
Epoch 60/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9033 - loss: 0.2492 - val_accu
racy: 0.7880 - val_loss: 0.5978
Epoch 61/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9140 - loss: 0.2264 - val_accu
racy: 0.8220 - val_loss: 0.6045
Epoch 62/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9007 - loss: 0.2483 - val_accu
racy: 0.8500 - val_loss: 0.4749
Epoch 63/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9193 - loss: 0.2190 - val_accu
racy: 0.8300 - val_loss: 0.5228
Epoch 64/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9100 - loss: 0.2196 - val_accu
racy: 0.8200 - val_loss: 0.6487
Epoch 65/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9113 - loss: 0.2286 - val_accu
racy: 0.8280 - val_loss: 0.5268
Epoch 66/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9133 - loss: 0.2369 - val_accu
racy: 0.8300 - val_loss: 0.4550
Epoch 67/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9100 - loss: 0.2174 - val_accu
racy: 0.8580 - val_loss: 0.4035
Epoch 68/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9260 - loss: 0.2032 - val_accu
racy: 0.7960 - val_loss: 0.6753
Epoch 69/100
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47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9193 - loss: 0.1914 - val_accu
racy: 0.8340 - val_loss: 0.7135
Epoch 70/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9180 - loss: 0.2062 - val_accu
racy: 0.8360 - val_loss: 0.4899
Epoch 71/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9113 - loss: 0.2036 - val_accu
racy: 0.8280 - val_loss: 0.5395
Epoch 72/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.9253 - loss: 0.1921 - val_accu
racy: 0.8560 - val_loss: 0.4522
Epoch 73/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9227 - loss: 0.2014 - val_accu
racy: 0.8340 - val_loss: 0.4902
Epoch 74/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9240 - loss: 0.1910 - val_accu
racy: 0.8440 - val_loss: 0.5030
Epoch 75/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9267 - loss: 0.1904 - val_accu
racy: 0.8240 - val_loss: 0.6820
Epoch 76/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9233 - loss: 0.1904 - val_accu
racy: 0.8380 - val_loss: 0.5241
Epoch 77/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9227 - loss: 0.2111 - val_accu
racy: 0.8340 - val_loss: 0.5535
Epoch 78/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9340 - loss: 0.1841 - val_accu
racy: 0.8180 - val_loss: 0.5372
Epoch 79/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9260 - loss: 0.1870 - val_accu
racy: 0.8260 - val_loss: 0.6314
Epoch 80/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9307 - loss: 0.1821 - val_accu
racy: 0.8220 - val_loss: 0.6323
Epoch 81/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9373 - loss: 0.1670 - val_accu
racy: 0.8460 - val_loss: 0.6907
Epoch 82/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9333 - loss: 0.1658 - val_accu
racy: 0.8500 - val_loss: 0.5878
Epoch 83/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9380 - loss: 0.1762 - val_accu
racy: 0.8360 - val_loss: 0.6309
Epoch 84/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9227 - loss: 0.1991 - val_accu
racy: 0.8580 - val_loss: 0.5556
Epoch 85/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9407 - loss: 0.1473 - val_accu
racy: 0.8460 - val_loss: 0.5624
Epoch 86/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9327 - loss: 0.1834 - val_accu
racy: 0.8600 - val_loss: 0.6011
Epoch 87/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9467 - loss: 0.1426 - val_accu
racy: 0.8360 - val_loss: 0.5991
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Epoch 88/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9500 - loss: 0.1442 - val_accu
racy: 0.8200 - val_loss: 0.5898
Epoch 89/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9487 - loss: 0.1535 - val_accu
racy: 0.8420 - val_loss: 0.5875
Epoch 90/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9373 - loss: 0.1521 - val_accu
racy: 0.8320 - val_loss: 0.6654
Epoch 91/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9520 - loss: 0.1281 - val_accu
racy: 0.8580 - val_loss: 0.6441
Epoch 92/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9247 - loss: 0.1987 - val_accu
racy: 0.8180 - val_loss: 0.6526
Epoch 93/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9487 - loss: 0.1358 - val_accu
racy: 0.7440 - val_loss: 1.7920
Epoch 94/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9500 - loss: 0.1526 - val_accu
racy: 0.8420 - val_loss: 0.6641
Epoch 95/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9547 - loss: 0.1217 - val_accu
racy: 0.8500 - val_loss: 0.6143
Epoch 96/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9547 - loss: 0.1414 - val_accu
racy: 0.8380 - val_loss: 0.7662
Epoch 97/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9407 - loss: 0.1656 - val_accu
racy: 0.8340 - val_loss: 0.6779
Epoch 98/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9453 - loss: 0.1478 - val_accu
racy: 0.8500 - val_loss: 0.6311
Epoch 99/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.9467 - loss: 0.1458 - val_accu
racy: 0.8380 - val_loss: 0.7803
Epoch 100/100
47/47 ━━━━━━━━━━━━━━━━━━━━ 2s 49ms/step - accuracy: 0.9440 - loss: 0.1471 - val_accu
racy: 0.8600 - val_loss: 0.7062
Epoch 1/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 63ms/step - accuracy: 0.5030 - loss: 0.6953 - val_accu
racy: 0.5000 - val_loss: 0.6941
Epoch 2/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.5015 - loss: 0.6939 - val_accu
racy: 0.5000 - val_loss: 0.6909
Epoch 3/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.5090 - loss: 0.6941 - val_accu
racy: 0.5320 - val_loss: 0.6827
Epoch 4/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.5675 - loss: 0.6853 - val_accu
racy: 0.6100 - val_loss: 0.6644
Epoch 5/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.5940 - loss: 0.6682 - val_accu
racy: 0.6400 - val_loss: 0.6550
Epoch 6/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.6430 - loss: 0.6456 - val_accu
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racy: 0.6580 - val_loss: 0.6015
Epoch 7/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.6560 - loss: 0.6287 - val_accu
racy: 0.6480 - val_loss: 0.6171
Epoch 8/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.6755 - loss: 0.6047 - val_accu
racy: 0.6880 - val_loss: 0.5859
Epoch 9/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.6665 - loss: 0.6078 - val_accu
racy: 0.6500 - val_loss: 0.6272
Epoch 10/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7025 - loss: 0.5848 - val_accu
racy: 0.6860 - val_loss: 0.5701
Epoch 11/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.7155 - loss: 0.5984 - val_accu
racy: 0.7060 - val_loss: 0.5529
Epoch 12/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7175 - loss: 0.5666 - val_accu
racy: 0.6940 - val_loss: 0.5554
Epoch 13/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7190 - loss: 0.5482 - val_accu
racy: 0.6980 - val_loss: 0.6271
Epoch 14/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7345 - loss: 0.5565 - val_accu
racy: 0.6980 - val_loss: 0.5915
Epoch 15/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7360 - loss: 0.5396 - val_accu
racy: 0.6720 - val_loss: 1.0833
Epoch 16/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7365 - loss: 0.5388 - val_accu
racy: 0.7540 - val_loss: 0.4969
Epoch 17/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7440 - loss: 0.5215 - val_accu
racy: 0.7420 - val_loss: 0.4973
Epoch 18/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7480 - loss: 0.5166 - val_accu
racy: 0.7760 - val_loss: 0.4973
Epoch 19/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7550 - loss: 0.5090 - val_accu
racy: 0.7420 - val_loss: 0.4855
Epoch 20/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7535 - loss: 0.5117 - val_accu
racy: 0.6540 - val_loss: 0.6841
Epoch 21/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7655 - loss: 0.4926 - val_accu
racy: 0.7500 - val_loss: 0.5051
Epoch 22/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.7765 - loss: 0.4818 - val_accu
racy: 0.7860 - val_loss: 0.4758
Epoch 23/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7775 - loss: 0.4622 - val_accu
racy: 0.7800 - val_loss: 0.5444
Epoch 24/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7765 - loss: 0.4645 - val_accu
racy: 0.7620 - val_loss: 0.4959
Epoch 25/100
```

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63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7700 - loss: 0.4611 - val_accu
racy: 0.7780 - val_loss: 0.4941
Epoch 26/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.7910 - loss: 0.4505 - val_accu
racy: 0.7880 - val_loss: 0.4806
Epoch 27/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.7905 - loss: 0.4426 - val_accu
racy: 0.8120 - val_loss: 0.4273
Epoch 28/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.7950 - loss: 0.4360 - val_accu
racy: 0.7840 - val_loss: 0.5061
Epoch 29/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 50ms/step - accuracy: 0.8100 - loss: 0.4298 - val_accu
racy: 0.8100 - val_loss: 0.4295
Epoch 30/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8095 - loss: 0.4090 - val_accu
racy: 0.8080 - val_loss: 0.4925
Epoch 31/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.8225 - loss: 0.3969 - val_accu
racy: 0.8080 - val_loss: 0.4241
Epoch 32/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8150 - loss: 0.4004 - val_accu
racy: 0.8040 - val_loss: 0.5079
Epoch 33/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.8300 - loss: 0.3714 - val_accu
racy: 0.8240 - val_loss: 0.4095
Epoch 34/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8360 - loss: 0.3802 - val_accu
racy: 0.7900 - val_loss: 0.5041
Epoch 35/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8400 - loss: 0.3663 - val_accu
racy: 0.8180 - val_loss: 0.4641
Epoch 36/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8370 - loss: 0.3691 - val_accu
racy: 0.8260 - val_loss: 0.4449
Epoch 37/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8390 - loss: 0.3709 - val_accu
racy: 0.8140 - val_loss: 0.5065
Epoch 38/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8460 - loss: 0.3481 - val_accu
racy: 0.8260 - val_loss: 0.4296
Epoch 39/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8540 - loss: 0.3392 - val_accu
racy: 0.7740 - val_loss: 0.6241
Epoch 40/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8450 - loss: 0.3522 - val_accu
racy: 0.7740 - val_loss: 0.5782
Epoch 41/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8535 - loss: 0.3304 - val_accu
racy: 0.8380 - val_loss: 0.4175
Epoch 42/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8580 - loss: 0.3141 - val_accu
racy: 0.7560 - val_loss: 0.6932
Epoch 43/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8675 - loss: 0.3169 - val_accu
racy: 0.8140 - val_loss: 0.4696
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Epoch 44/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.8655 - loss: 0.3203 - val_accu
racy: 0.8120 - val_loss: 0.4462
Epoch 45/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8770 - loss: 0.2840 - val_accu
racy: 0.8300 - val_loss: 0.4856
Epoch 46/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8795 - loss: 0.2959 - val_accu
racy: 0.8140 - val_loss: 0.5544
Epoch 47/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 49ms/step - accuracy: 0.8775 - loss: 0.3025 - val_accu
racy: 0.8540 - val_loss: 0.3751
Epoch 48/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8865 - loss: 0.2750 - val_accu
racy: 0.8380 - val_loss: 0.4550
Epoch 49/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8860 - loss: 0.2819 - val_accu
racy: 0.8460 - val_loss: 0.3853
Epoch 50/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8895 - loss: 0.2742 - val_accu
racy: 0.8380 - val_loss: 0.4574
Epoch 51/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8965 - loss: 0.2585 - val_accu
racy: 0.8320 - val_loss: 0.5477
Epoch 52/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8925 - loss: 0.2599 - val_accu
racy: 0.8600 - val_loss: 0.4185
Epoch 53/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.8860 - loss: 0.2730 - val_accu
racy: 0.8140 - val_loss: 0.5264
Epoch 54/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9025 - loss: 0.2351 - val_accu
racy: 0.6900 - val_loss: 1.0361
Epoch 55/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9010 - loss: 0.2504 - val_accu
racy: 0.8480 - val_loss: 0.5136
Epoch 56/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.8930 - loss: 0.2651 - val_accu
racy: 0.8420 - val_loss: 0.4505
Epoch 57/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.8985 - loss: 0.2517 - val_accu
racy: 0.8420 - val_loss: 0.5176
Epoch 58/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9065 - loss: 0.2280 - val_accu
racy: 0.8380 - val_loss: 0.4917
Epoch 59/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.9090 - loss: 0.2298 - val_accu
racy: 0.8320 - val_loss: 0.5929
Epoch 60/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9175 - loss: 0.2241 - val_accu
racy: 0.8380 - val_loss: 0.6101
Epoch 61/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.9075 - loss: 0.2219 - val_accu
racy: 0.8320 - val_loss: 0.5232
Epoch 62/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.9130 - loss: 0.2114 - val_accu
```

```
racy: 0.7940 - val_loss: 0.6096
Epoch 63/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.9120 - loss: 0.1979 - val_accu
racy: 0.8240 - val_loss: 0.5855
Epoch 64/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 45ms/step - accuracy: 0.9255 - loss: 0.1986 - val_accu
racy: 0.7580 - val_loss: 0.9670
Epoch 65/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 54ms/step - accuracy: 0.9100 - loss: 0.2114 - val_accu
racy: 0.8440 - val_loss: 0.5539
Epoch 66/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 19s 313ms/step - accuracy: 0.9280 - loss: 0.1849 - val_ac
curacy: 0.8180 - val_loss: 0.7470
Epoch 67/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9265 - loss: 0.1975 - val_accu
racy: 0.8520 - val_loss: 0.4972
Epoch 68/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9140 - loss: 0.2105 - val_accu
racy: 0.8380 - val_loss: 0.6254
Epoch 69/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9320 - loss: 0.1853 - val_accu
racy: 0.8400 - val_loss: 0.6005
Epoch 70/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9270 - loss: 0.1916 - val_accu
racy: 0.8400 - val_loss: 0.5294
Epoch 71/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9215 - loss: 0.1926 - val_accu
racy: 0.8300 - val_loss: 0.5626
Epoch 72/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 50ms/step - accuracy: 0.9305 - loss: 0.1725 - val_accu
racy: 0.8460 - val_loss: 0.6636
Epoch 73/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 50ms/step - accuracy: 0.9175 - loss: 0.2058 - val_accu
racy: 0.8240 - val_loss: 0.5940
Epoch 74/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 50ms/step - accuracy: 0.9340 - loss: 0.1753 - val_accu
racy: 0.8540 - val_loss: 0.5680
Epoch 75/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 50ms/step - accuracy: 0.9370 - loss: 0.1650 - val_accu
racy: 0.8400 - val_loss: 0.7872
Epoch 76/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9360 - loss: 0.1802 - val_accu
racy: 0.8420 - val_loss: 0.6248
Epoch 77/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9305 - loss: 0.1763 - val_accu
racy: 0.8800 - val_loss: 0.5114
Epoch 78/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9365 - loss: 0.1640 - val_accu
racy: 0.8380 - val_loss: 0.6188
Epoch 79/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9400 - loss: 0.1704 - val_accu
racy: 0.8200 - val_loss: 0.8033
Epoch 80/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9355 - loss: 0.1689 - val_accu
racy: 0.8340 - val_loss: 0.6569
Epoch 81/100
```

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63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9390 - loss: 0.1762 - val_accu
racy: 0.8760 - val_loss: 0.5383
Epoch 82/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9400 - loss: 0.1740 - val_accu
racy: 0.8480 - val_loss: 0.4885
Epoch 83/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9430 - loss: 0.1677 - val_accu
racy: 0.8500 - val_loss: 0.8982
Epoch 84/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9430 - loss: 0.1582 - val_accu
racy: 0.8680 - val_loss: 0.4929
Epoch 85/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9320 - loss: 0.1696 - val_accu
racy: 0.8480 - val_loss: 0.6893
Epoch 86/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 46ms/step - accuracy: 0.9410 - loss: 0.1622 - val_accu
racy: 0.8580 - val_loss: 0.5366
Epoch 87/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9475 - loss: 0.1437 - val_accu
racy: 0.8480 - val_loss: 0.6474
Epoch 88/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9460 - loss: 0.1581 - val_accu
racy: 0.8240 - val_loss: 0.7293
Epoch 89/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9460 - loss: 0.1796 - val_accu
racy: 0.8580 - val_loss: 0.6144
Epoch 90/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9385 - loss: 0.1889 - val_accu
racy: 0.8600 - val_loss: 0.6093
Epoch 91/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9420 - loss: 0.1517 - val_accu
racy: 0.8400 - val_loss: 0.5370
Epoch 92/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 49ms/step - accuracy: 0.9455 - loss: 0.1490 - val_accu
racy: 0.7780 - val_loss: 1.0957
Epoch 93/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 47ms/step - accuracy: 0.9475 - loss: 0.1647 - val_accu
racy: 0.8680 - val_loss: 0.5993
Epoch 94/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9450 - loss: 0.1733 - val_accu
racy: 0.8160 - val_loss: 0.5836
Epoch 95/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9510 - loss: 0.1440 - val_accu
racy: 0.8520 - val_loss: 0.7011
Epoch 96/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9445 - loss: 0.1577 - val_accu
racy: 0.8680 - val_loss: 0.5361
Epoch 97/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9470 - loss: 0.1424 - val_accu
racy: 0.8300 - val_loss: 0.8169
Epoch 98/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9470 - loss: 0.1444 - val_accu
racy: 0.8740 - val_loss: 0.6025
Epoch 99/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9480 - loss: 0.1515 - val_accu
racy: 0.8760 - val_loss: 0.4840
```

```
Epoch 100/100
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 48ms/step - accuracy: 0.9470 - loss: 0.1505 - val_accu
racy: 0.8680 - val_loss: 0.5146
10/16 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7500 - loss: 0.5318
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
16/16 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7460 - loss: 0.5466
1000 Test accuracy: 0.746
16/16 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.8140 - loss: 0.4561
1500 Test accuracy: 0.814
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
16/16 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.8260 - loss: 0.4190
2000 Test accuracy: 0.826
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
```

In [18]:

```python
############PRETRAINED model function
def make_pretrained_model():

    conv_base = keras.applications.vgg16.VGG16(
        weights="imagenet",
        include_top=False,
        input_shape=(180, 180, 3)
    )

    conv_base.trainable = False

    data_augmentation = keras.Sequential(
        [
            layers.RandomFlip("horizontal"),
            layers.RandomRotation(0.1),
            layers.RandomZoom(0.2),
        ]
    )

    inputs = keras.Input(shape=(180, 180, 3))

    x = data_augmentation(inputs)
    x = keras.applications.vgg16.preprocess_input(x)

    x = conv_base(x)
    x = layers.Flatten()(x)
    x = layers.Dense(256, activation="relu")(x)
    x = layers.Dropout(0.5)(x)

    outputs = layers.Dense(1, activation="sigmoid")(x)

    model = keras.Model(inputs, outputs)

    model.compile(
        loss="binary_crossentropy",
        optimizer="rmsprop",
        metrics=["accuracy"]
    )

    return model
```

In [20]:
```
#make three pretrained models
pretrained_1000 = make_pretrained_model()
pretrained_1500 = make_pretrained_model()
pretrained_2000 = make_pretrained_model()
#function for pretrained callbacks
def pretrained_callbacks(training_size):
    return [
        keras.callbacks.ModelCheckpoint(
            filepath=f"convnet_pretrained_{training_size}.keras",
            save_best_only=True,
            monitor="val_loss"
        )
    ]
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/v
gg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5
58889256/58889256 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
```

In [21]:
```
#fit and do callbacks
phistory_1000 = pretrained_1000.fit(
    train_1000_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=pretrained_callbacks("1000")
)

phistory_1500 = pretrained_1500.fit(
    train_1500_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=pretrained_callbacks("1500")
)

phistory_2000 = pretrained_2000.fit(
    train_2000_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=pretrained_callbacks("2000")
)


pretrained_model_1000 = keras.models.load_model(
    "convnet_pretrained_1000.keras")
pret_loss_1000, pret_acc_1000 = pretrained_model_1000.evaluate(test_dataset)
print(f"1000 Test accuracy: {pret_acc_1000:.3f}")

pretrained_model_1500 = keras.models.load_model(
    "convnet_pretrained_1500.keras")
pret_loss_1500, pret_acc_1500 = pretrained_model_1500.evaluate(test_dataset)
print(f"1500 Test accuracy: {pret_acc_1500:.3f}")

pretrained_model_2000 = keras.models.load_model(
    "convnet_pretrained_2000.keras")
pret_loss_2000, pret_acc_2000 = pretrained_model_2000.evaluate(test_dataset)
print(f"2000 Test accuracy: {pret_acc_2000:.3f}")
```

```
Epoch 1/30
```

```
2026-10-15 19:21:49.018318: W external/local_xla/xla/tsl/framework/bfc_allocator.cc:
310] Allocator (GPU_0_bfc) ran out of memory trying to allocate 3.04GiB with freed_b
y_count=0. The caller indicates that this is not a failure, but this may mean that t
here could be performance gains if more memory were available.
2026-10-15 19:21:49.823710: W external/local_xla/xla/tsl/framework/bfc_allocator.cc:
310] Allocator (GPU_0_bfc) ran out of memory trying to allocate 2.55GiB with freed_b
y_count=0. The caller indicates that this is not a failure, but this may mean that t
here could be performance gains if more memory were available.
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 0s 134ms/step - accuracy: 0.8580 - loss: 10.5177
```

```
2026-10-15 19:21:57.400154: W external/local_xla/xla/tsl/framework/bfc_allocator.cc:
310] Allocator (GPU_0_bfc) ran out of memory trying to allocate 2.66GiB with freed_b
y_count=0. The caller indicates that this is not a failure, but this may mean that t
here could be performance gains if more memory were available.
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 12s 256ms/step - accuracy: 0.8580 - loss: 10.5177 - val_a
ccuracy: 0.9460 - val_loss: 0.9273
Epoch 2/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 164ms/step - accuracy: 0.9150 - loss: 1.7837 - val_acc
uracy: 0.9780 - val_loss: 0.4234
Epoch 3/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 155ms/step - accuracy: 0.9470 - loss: 0.8219 - val_acc
uracy: 0.9620 - val_loss: 0.7019
Epoch 4/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 163ms/step - accuracy: 0.9430 - loss: 0.6666 - val_acc
uracy: 0.9720 - val_loss: 0.2580
Epoch 5/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 155ms/step - accuracy: 0.9550 - loss: 0.5532 - val_acc
uracy: 0.9720 - val_loss: 0.2924
Epoch 6/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 163ms/step - accuracy: 0.9590 - loss: 0.3437 - val_acc
uracy: 0.9820 - val_loss: 0.1569
Epoch 7/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9600 - loss: 0.2556 - val_acc
uracy: 0.9780 - val_loss: 0.2301
Epoch 8/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9770 - loss: 0.1272 - val_acc
uracy: 0.9860 - val_loss: 0.1860
Epoch 9/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9730 - loss: 0.2709 - val_acc
uracy: 0.9820 - val_loss: 0.2305
Epoch 10/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9740 - loss: 0.1886 - val_acc
uracy: 0.9820 - val_loss: 0.1747
Epoch 11/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9830 - loss: 0.2143 - val_acc
uracy: 0.9740 - val_loss: 0.3119
Epoch 12/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9800 - loss: 0.1738 - val_acc
uracy: 0.9700 - val_loss: 0.3870
Epoch 13/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9720 - loss: 0.2917 - val_acc
uracy: 0.9740 - val_loss: 0.3153
Epoch 14/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 158ms/step - accuracy: 0.9740 - loss: 0.2733 - val_acc
uracy: 0.9740 - val_loss: 0.2207
Epoch 15/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 156ms/step - accuracy: 0.9800 - loss: 0.1484 - val_acc
uracy: 0.9780 - val_loss: 0.3162
Epoch 16/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9840 - loss: 0.1284 - val_acc
uracy: 0.9780 - val_loss: 0.2411
Epoch 17/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9840 - loss: 0.1314 - val_acc
uracy: 0.9800 - val_loss: 0.2121
Epoch 18/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9910 - loss: 0.0415 - val_acc
uracy: 0.9800 - val_loss: 0.2220
Epoch 19/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9910 - loss: 0.0583 - val_acc
uracy: 0.9800 - val_loss: 0.2626
```

```
Epoch 20/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 166ms/step - accuracy: 0.9850 - loss: 0.1321 - val_acc
uracy: 0.9840 - val_loss: 0.1525
Epoch 21/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 158ms/step - accuracy: 0.9860 - loss: 0.1543 - val_acc
uracy: 0.9740 - val_loss: 0.3045
Epoch 22/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 159ms/step - accuracy: 0.9880 - loss: 0.0921 - val_acc
uracy: 0.9780 - val_loss: 0.2662
Epoch 23/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 158ms/step - accuracy: 0.9920 - loss: 0.0661 - val_acc
uracy: 0.9840 - val_loss: 0.1843
Epoch 24/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9850 - loss: 0.1395 - val_acc
uracy: 0.9840 - val_loss: 0.2277
Epoch 25/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 165ms/step - accuracy: 0.9910 - loss: 0.1070 - val_acc
uracy: 0.9900 - val_loss: 0.1465
Epoch 26/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9890 - loss: 0.1417 - val_acc
uracy: 0.9840 - val_loss: 0.1850
Epoch 27/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 158ms/step - accuracy: 0.9880 - loss: 0.0885 - val_acc
uracy: 0.9820 - val_loss: 0.1615
Epoch 28/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 158ms/step - accuracy: 0.9890 - loss: 0.0766 - val_acc
uracy: 0.9860 - val_loss: 0.1797
Epoch 29/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9940 - loss: 0.0281 - val_acc
uracy: 0.9860 - val_loss: 0.1595
Epoch 30/30
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 157ms/step - accuracy: 0.9920 - loss: 0.0336 - val_acc
uracy: 0.9820 - val_loss: 0.1758
Epoch 1/30
46/47 ━━━━━━━━━━━━━━━━━━━━ 0s 107ms/step - accuracy: 0.8716 - loss: 7.0833
2026-10-15 19:24:31.066599: W external/local_xla/xla/tsl/framework/bfc_allocator.cc:
310] Allocator (GPU_0_bfc) ran out of memory trying to allocate 2.91GiB with freed_b
y_count=0. The caller indicates that this is not a failure, but this may mean that t
here could be performance gains if more memory were available.
```

```
47/47 ━━━━━━━━━━━━━━━━━━━━ 10s 204ms/step - accuracy: 0.8727 - loss: 6.9720 - val_ac
curacy: 0.9780 - val_loss: 0.3204
Epoch 2/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9387 - loss: 0.7958 - val_acc
uracy: 0.9740 - val_loss: 0.4581
Epoch 3/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 148ms/step - accuracy: 0.9460 - loss: 0.5512 - val_acc
uracy: 0.9720 - val_loss: 0.3170
Epoch 4/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9507 - loss: 0.4689 - val_acc
uracy: 0.9720 - val_loss: 0.3558
Epoch 5/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 142ms/step - accuracy: 0.9613 - loss: 0.3048 - val_acc
uracy: 0.9660 - val_loss: 0.3219
Epoch 6/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 150ms/step - accuracy: 0.9613 - loss: 0.3060 - val_acc
uracy: 0.9740 - val_loss: 0.2610
Epoch 7/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 149ms/step - accuracy: 0.9647 - loss: 0.3711 - val_acc
uracy: 0.9820 - val_loss: 0.1651
Epoch 8/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 148ms/step - accuracy: 0.9667 - loss: 0.2932 - val_acc
uracy: 0.9900 - val_loss: 0.0874
Epoch 9/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9727 - loss: 0.1953 - val_acc
uracy: 0.9820 - val_loss: 0.1143
Epoch 10/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9767 - loss: 0.1354 - val_acc
uracy: 0.9840 - val_loss: 0.1351
Epoch 11/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 149ms/step - accuracy: 0.9753 - loss: 0.1950 - val_acc
uracy: 0.9880 - val_loss: 0.0684
Epoch 12/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - accuracy: 0.9813 - loss: 0.1111 - val_acc
uracy: 0.9800 - val_loss: 0.1945
Epoch 13/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 145ms/step - accuracy: 0.9753 - loss: 0.1693 - val_acc
uracy: 0.9900 - val_loss: 0.0873
Epoch 14/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - accuracy: 0.9767 - loss: 0.1517 - val_acc
uracy: 0.9900 - val_loss: 0.0803
Epoch 15/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9760 - loss: 0.1859 - val_acc
uracy: 0.9740 - val_loss: 0.2311
Epoch 16/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9827 - loss: 0.0894 - val_acc
uracy: 0.9840 - val_loss: 0.0873
Epoch 17/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - accuracy: 0.9767 - loss: 0.1685 - val_acc
uracy: 0.9760 - val_loss: 0.2191
Epoch 18/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - accuracy: 0.9893 - loss: 0.0758 - val_acc
uracy: 0.9720 - val_loss: 0.2263
Epoch 19/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9793 - loss: 0.1397 - val_acc
uracy: 0.9820 - val_loss: 0.2199
```

```
Epoch 20/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - accuracy: 0.9873 - loss: 0.0732 - val_acc
uracy: 0.9800 - val_loss: 0.1607
Epoch 21/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 145ms/step - accuracy: 0.9873 - loss: 0.0996 - val_acc
uracy: 0.9860 - val_loss: 0.1259
Epoch 22/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 147ms/step - accuracy: 0.9840 - loss: 0.1315 - val_acc
uracy: 0.9860 - val_loss: 0.1101
Epoch 23/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 142ms/step - accuracy: 0.9887 - loss: 0.0583 - val_acc
uracy: 0.9780 - val_loss: 0.1643
Epoch 24/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 142ms/step - accuracy: 0.9900 - loss: 0.0669 - val_acc
uracy: 0.9720 - val_loss: 0.2192
Epoch 25/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 141ms/step - accuracy: 0.9900 - loss: 0.0522 - val_acc
uracy: 0.9760 - val_loss: 0.3010
Epoch 26/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 141ms/step - accuracy: 0.9887 - loss: 0.0742 - val_acc
uracy: 0.9820 - val_loss: 0.1890
Epoch 27/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 141ms/step - accuracy: 0.9920 - loss: 0.0527 - val_acc
uracy: 0.9840 - val_loss: 0.1817
Epoch 28/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 141ms/step - accuracy: 0.9913 - loss: 0.0584 - val_acc
uracy: 0.9720 - val_loss: 0.3632
Epoch 29/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 141ms/step - accuracy: 0.9887 - loss: 0.0824 - val_acc
uracy: 0.9880 - val_loss: 0.1736
Epoch 30/30
47/47 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - accuracy: 0.9873 - loss: 0.0818 - val_acc
uracy: 0.9780 - val_loss: 0.1234
Epoch 1/30
62/63 ━━━━━━━━━━━━━━━━━━━━ 0s 106ms/step - accuracy: 0.8805 - loss: 6.3980
2026-10-15 19:27:58.731278: W external/local_xla/xla/tsl/framework/bfc_allocator.cc:
310] Allocator (GPU_0_bfc) ran out of memory trying to allocate 2.54GiB with freed_b
y_count=0. The caller indicates that this is not a failure, but this may mean that t
here could be performance gains if more memory were available.
```

```
63/63 ━━━━━━━━━━━━━━━━━━━━ 11s 162ms/step - accuracy: 0.8815 - loss: 6.3468 - val_ac
curacy: 0.9620 - val_loss: 0.7496
Epoch 2/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 137ms/step - accuracy: 0.9345 - loss: 0.9293 - val_acc
uracy: 0.9740 - val_loss: 0.2682
Epoch 3/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 135ms/step - accuracy: 0.9475 - loss: 0.5132 - val_acc
uracy: 0.9860 - val_loss: 0.1218
Epoch 4/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9510 - loss: 0.3702 - val_acc
uracy: 0.9680 - val_loss: 0.3254
Epoch 5/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9640 - loss: 0.3500 - val_acc
uracy: 0.9780 - val_loss: 0.2967
Epoch 6/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9560 - loss: 0.4581 - val_acc
uracy: 0.9780 - val_loss: 0.2133
Epoch 7/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9680 - loss: 0.2733 - val_acc
uracy: 0.9780 - val_loss: 0.1320
Epoch 8/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9650 - loss: 0.2639 - val_acc
uracy: 0.9780 - val_loss: 0.2288
Epoch 9/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9715 - loss: 0.2386 - val_acc
uracy: 0.9820 - val_loss: 0.1626
Epoch 10/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9700 - loss: 0.2315 - val_acc
uracy: 0.9780 - val_loss: 0.2473
Epoch 11/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 132ms/step - accuracy: 0.9690 - loss: 0.2379 - val_acc
uracy: 0.9860 - val_loss: 0.1307
Epoch 12/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 133ms/step - accuracy: 0.9770 - loss: 0.1379 - val_acc
uracy: 0.9800 - val_loss: 0.1624
Epoch 13/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 131ms/step - accuracy: 0.9790 - loss: 0.1759 - val_acc
uracy: 0.9860 - val_loss: 0.1660
Epoch 14/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 136ms/step - accuracy: 0.9810 - loss: 0.1795 - val_acc
uracy: 0.9860 - val_loss: 0.0969
Epoch 15/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 135ms/step - accuracy: 0.9850 - loss: 0.1248 - val_acc
uracy: 0.9880 - val_loss: 0.0842
Epoch 16/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 133ms/step - accuracy: 0.9895 - loss: 0.0828 - val_acc
uracy: 0.9820 - val_loss: 0.1439
Epoch 17/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 135ms/step - accuracy: 0.9825 - loss: 0.1496 - val_acc
uracy: 0.9900 - val_loss: 0.0872
Epoch 18/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 138ms/step - accuracy: 0.9830 - loss: 0.1323 - val_acc
uracy: 0.9880 - val_loss: 0.0721
Epoch 19/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 134ms/step - accuracy: 0.9835 - loss: 0.1180 - val_acc
uracy: 0.9860 - val_loss: 0.0783
```

```
Epoch 20/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 133ms/step - accuracy: 0.9855 - loss: 0.0808 - val_acc
uracy: 0.9840 - val_loss: 0.1396
Epoch 21/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 133ms/step - accuracy: 0.9840 - loss: 0.1213 - val_acc
uracy: 0.9860 - val_loss: 0.1899
Epoch 22/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 136ms/step - accuracy: 0.9905 - loss: 0.0597 - val_acc
uracy: 0.9840 - val_loss: 0.1119
Epoch 23/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 140ms/step - accuracy: 0.9825 - loss: 0.1396 - val_acc
uracy: 0.9840 - val_loss: 0.1587
Epoch 24/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 139ms/step - accuracy: 0.9855 - loss: 0.1094 - val_acc
uracy: 0.9860 - val_loss: 0.0707
Epoch 25/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 135ms/step - accuracy: 0.9920 - loss: 0.0321 - val_acc
uracy: 0.9840 - val_loss: 0.1931
Epoch 26/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 134ms/step - accuracy: 0.9865 - loss: 0.0933 - val_acc
uracy: 0.9780 - val_loss: 0.2130
Epoch 27/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 134ms/step - accuracy: 0.9945 - loss: 0.0333 - val_acc
uracy: 0.9900 - val_loss: 0.1222
Epoch 28/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 133ms/step - accuracy: 0.9895 - loss: 0.0515 - val_acc
uracy: 0.9740 - val_loss: 0.1620
Epoch 29/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 134ms/step - accuracy: 0.9855 - loss: 0.0979 - val_acc
uracy: 0.9820 - val_loss: 0.2061
Epoch 30/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 137ms/step - accuracy: 0.9890 - loss: 0.0807 - val_acc
uracy: 0.9900 - val_loss: 0.0627
 9/16 ━━━━━━━━━━━━━━━━━━━━ 0s 103ms/step - accuracy: 0.9757 - loss: 0.6723
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
16/16 ━━━━━━━━━━━━━━━━━━━━ 2s 102ms/step - accuracy: 0.9780 - loss: 0.6375
1000 Test accuracy: 0.978
 4/16 ━━━━━━━━━━━━━━━━━━━━ 1s 104ms/step - accuracy: 0.9688 - loss: 0.1555
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
16/16 ━━━━━━━━━━━━━━━━━━━━ 2s 101ms/step - accuracy: 0.9700 - loss: 0.3069
1500 Test accuracy: 0.970
 4/16 ━━━━━━━━━━━━━━━━━━━━ 1s 102ms/step - accuracy: 0.9766 - loss: 0.4720
Corrupt JPEG data: 128 extraneous bytes before marker 0xd9
16/16 ━━━━━━━━━━━━━━━━━━━━ 2s 101ms/step - accuracy: 0.9760 - loss: 0.4201
2000 Test accuracy: 0.976
```

In [22]:

```python
import numpy as np

#this code stores variables to show the best epoch for each model
#scratch models

best_epoch_1000 = np.argmin(history_1000.history["val_loss"]) + 1
best_val_acc_1000 = history_1000.history["val_accuracy"][best_epoch_1000 - 1]

best_epoch_1500 = np.argmin(history_1500.history["val_loss"]) + 1
```

```
best_val_acc_1500 = history_1500.history["val_accuracy"][best_epoch_1500 - 1]

best_epoch_2000 = np.argmin(history_2000.history["val_loss"]) + 1
best_val_acc_2000 = history_2000.history["val_accuracy"][best_epoch_2000 - 1]


#Pretrained models

best_p_epoch_1000 = np.argmin(phistory_1000.history["val_loss"]) + 1
best_p_val_acc_1000 = phistory_1000.history["val_accuracy"][best_p_epoch_1000 - 1]

best_p_epoch_1500 = np.argmin(phistory_1500.history["val_loss"]) + 1
best_p_val_acc_1500 = phistory_1500.history["val_accuracy"][best_p_epoch_1500 - 1]

best_p_epoch_2000 = np.argmin(phistory_2000.history["val_loss"]) + 1
best_p_val_acc_2000 = phistory_2000.history["val_accuracy"][best_p_epoch_2000 - 1]
```

In [26]:

```
#Tabular Final Results – EXECUTE LAST- then copy paste the table up top
%pip install pandas
import pandas as pd
from IPython.display import display

results_table = pd.DataFrame(
    {
        "Model type": [
            "From scratch",
            "From scratch",
            "From scratch",
            "Pretrained VGG16",
            "Pretrained VGG16",
            "Pretrained VGG16",
        ],
        "Training images": [
            1000,
            1500,
            2000,
            1000,
            1500,
            2000,
        ],
        "Best epoch": [
            best_epoch_1000,
            best_epoch_1500,
            best_epoch_2000,
            best_p_epoch_1000,
            best_p_epoch_1500,
            best_p_epoch_2000,
        ],
        "Validation accuracy at best epoch": [
            best_val_acc_1000,
            best_val_acc_1500,
            best_val_acc_2000,
            best_p_val_acc_1000,
            best_p_val_acc_1500,
            best_p_val_acc_2000,
        ],
```

```
        "Test accuracy": [
            test_acc_1000,
            test_acc_1500,
            test_acc_2000,
            pret_acc_1000,
            pret_acc_1500,
            pret_acc_2000,
        ],
    }
)

display(
    results_table.style.format(
        {
            "Validation accuracy at best epoch": "{:.3%}",
            "Test accuracy": "{:.3%}",
        }
    ))
```

```
Collecting pandas
  Downloading pandas-3.0.6-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)
Requirement already satisfied: numpy>=1.26.0 in /home/claxta11/miniconda3/envs/tensorflow/lib/python3.12/site-packages (from pandas) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /home/claxta11/miniconda3/envs/tensorflow/lib/python3.12/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /home/claxta11/miniconda3/envs/tensorflow/lib/python3.12/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading pandas-3.0.6-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (10.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.8/10.8 MB 37.7 MB/s  0:00:00eta 0:00:01
Installing collected packages: pandas
Successfully installed pandas-3.0.6
Note: you may need to restart the kernel to use updated packages.
```

| | Model type | Training images | Best epoch | Validation accuracy at best epoch | Test accuracy |
|---|---|---|---|---|---|
| **0** | From scratch | 1000 | 38 | 75.800% | 74.600% |
| **1** | From scratch | 1500 | 46 | 82.400% | 81.400% |
| **2** | From scratch | 2000 | 47 | 85.400% | 82.600% |
| **3** | Pretrained VGG16 | 1000 | 25 | 99.000% | 97.800% |
| **4** | Pretrained VGG16 | 1500 | 11 | 98.800% | 97.000% |
| **5** | Pretrained VGG16 | 2000 | 30 | 99.000% | 97.600% |

In [ ]: